Newton's Law of Cooling

Newton formulated the principle that the rate of change of the temperature of an object is proportional to the difference between the object's temperature and the temperature of the surroundings.

Suppose that the temperature of an object at time t is given by $T(t)$. Denote the surrounding temperature T^*, and assume it is constant. Write down a differential equation expressing Newton's Law of Cooling:

$$\frac{dT}{dt} = _____$$

Note that if the surrounding temperature is greater than the temperature of the object, we expect the temperature of the object to _____, whereas if the surrounding temperature is lower than the temperature of the object, we expect the temperature of the object to _____. Given that we usually take our constants of proportionality to be positive, does your equation above reflect this? If not, correct it.

What is the equilibrium solution to your differential equation? By sketching a slopefield (pick values of k and T^* if needed), determine if the equilibrium is stable or unstable.

Plot two solutions of the equation on your slopefield. One with $T(0) > T^*$, and one with $T(0) < T^*$. What is the behavior of the solutions as $t \to \infty$? Explain why this makes physical sense.

Heating and Cooling Problems

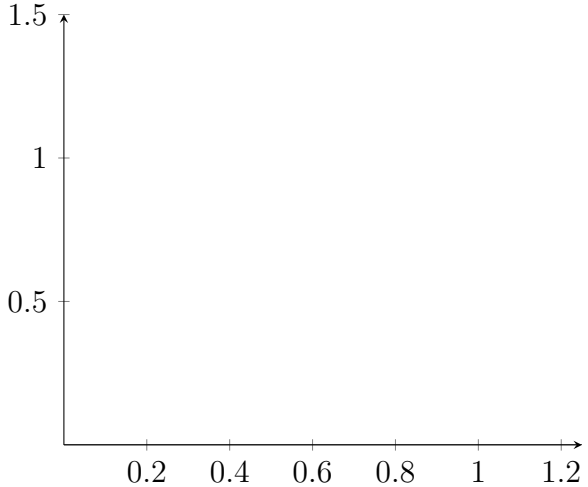
1. □Regrade Suppose that a cup of coffee, which has temperature 110°F, is placed in a room which is at temperature 72°F. Suppose that we measure time in hours after the placement of the coffee in the room and that the constant of proportionality is 0.7 per hour. By writing down and solving an initial value problem, find how long will it take for the temperature of the cup of coffee to reach 90°F?

2. □Regrade Suppose that the cup of coffee in problem 1 has temperature 40°F at $t = 0$ since it was mistakenly placed in the refrigerator. Find a formula for the temperature as a function of time.

3. □Regrade An ingot of iron ore at 1000°F is plunged into a water bath whose temperature is kept at 60°F. The temperature of the ingot decreases according to Newton's Law of Cooling with constant of proportionality $k = 3$ when time is measured in minutes. Find a formula for the temperature of the ingot as a function of time. How long will it take until the temperature reaches 70°F?

Understanding k

Draw the graphs of e^{-x}, e^{-2x}, and e^{-3x} on the axes below for $x \geq 0$.



Fill in the blank: In the function e^{-kx}, the larger k is, the _____ the function decays.

Thinking back to the Growth and Decay lab, consider the following description: a radioactive substance is decaying at a rate of 5% per day. Suppose we begin with N_0 moles of it. Write an initial value problem for this.

Newton's Law of Cooling is very similar:

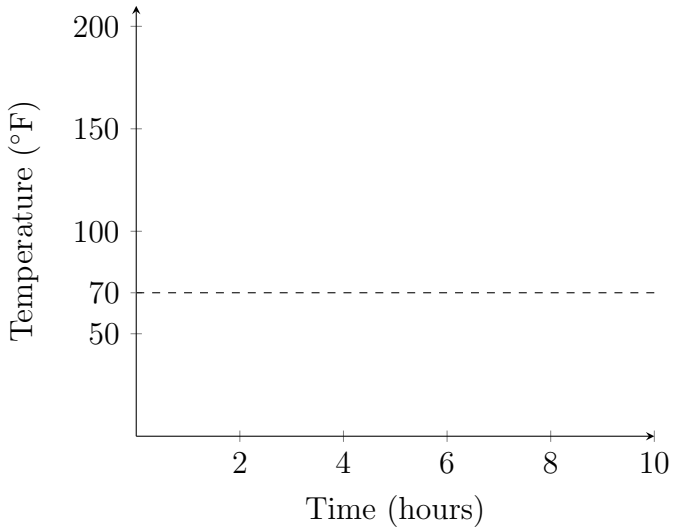
$$\frac{dT}{dt} = -k(T - T^*)$$

We can therefore view k as a rate of decay here as well.

4. By separating variables as before, show that the general solution to the differential □Regrade equation above is

$$T(t) = Ce^{-kt} + T^*.$$

5. Consider two situations in which an object with initial temperature 200°F is placed □Regrade in a room with constant temperature 70°F. In one situation, $k = 0.5$. In the other, $k = 3$. Find C and draw the graphs of the two objects' temperatures below. Then find the objects' temperatures at $t = 1$ hour to the nearest degree F.



6. ☐Regrade (a) In the previous problem, both objects were cups of tea. One cup was open and made of metal. The other was an insulated reusable take home cup. Which was which? Explain your answer carefully.

(b) What are the units of k in this situation? Explain.

7. Fill in all the blanks in the summary and cross out the incorrect choices.

☐Regrade

Newton's Law of Cooling

If an object is placed in a room with constant temperature T^*, its temperature changes in proportion to the difference between its current temperature $T(t)$ and the ambient temperature:

$$\frac{dT}{dt} = _____.$$

There is a _____ equilibrium at $T =$ _____.

That is, as time passes, the temperature of the object _____.

If the initial temperature of the object is T_0, the solution to this equation is

$$T(t) = _____.$$

k is a measure of heat conductivity: If k is larger, the temperature changes faster/slower. Therefore k is large when the object conducts heat well/less well.